CAS (Heimdall) – system functional description

Outlet system

ScaleAQ Automation Document CAS-FD-30, rev. A1

November 5, 2026

Abstract

The bottom outlet of the CAS closed pen has four dampers. Each damper is moved by a hydraulic cylinder. A Servi hydraulic power unit (HPU) with a four-section HAWE PSL valve bank drives the cylinders. The CAS PLC sets all four dampers to a common opening per operating mode. It finds the position of each cylinder by counting pulses from an oil-flow meter, and detects end positions by a rise in line pressure. This document describes the equipment, the control principle, the sequences, interlocks and faults, and lists what is still open. It describes the PLC function as implemented in `FB.Outlet.OutletControl`; it does not redesign it.

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1 Introduction

1.1 Purpose

This document is the system functional description of the outlet system of the CAS (Heimdall) closed fish pen. It is written for engineers, operators, the project team and the suppliers (Østerbø Solutions, Servi). After reading it, the reader should know:

- which instruments and actuators the outlet has, and where their signals go;
- how the PLC finds the damper position and moves the dampers;
- what happens on start-up, stop, faults, emergency stop and power loss;
- which questions are still open.

The structure follows Cooling's system functional specification [COO, §14.3]: block diagram description (§14.3.2, Section 2), functional diagram description (§14.3.3, Section 3), and the requirements specification, functional (§14.3.4a, Section 4) and non-functional (§14.3.4b, Section 5).

1.2 Scope

In scope: the four outlet dampers, the four cylinders, the HPU and valve cabinet (Servi order S6377070), junction box JB1 (Servi drawing 18068) with its local operation panel, the signals to the CAS remote I/O (CommonRIO), the PLC function block `FB.Outlet.OutletControl`, and the operator functions on the HMI.

Out of scope: the dead fish pump (DC-81) and sludge pump (SS-91) at the outlet, which Barge Control controls; the level control LIC01, which uses the operating mode set here but is described in the inlet FD; the mechanical design of the dampers (Østerbø).

1.3 Sources

ID	Document
[COO]	J. Cooling, *Software Engineering for Real-Time Systems*, 2020, §14.3 System functional specifications
[FD]	P-15169-CAS-FD-01 Funksjonsbeskrivelse CAS, rev. B, §3.2.3 Utløpsspjeld (Norwegian)
[AL]	P-15169-CAS-FD-02 Alarmliste CAS Heimdall, rev. A
[A1]	Servi 2939238 rev. A1, Hydraulikk skjema, spjeld til utløp HY, art. 810936, 18.06.2026 (current)
[T1]	Servi 2799175 rev. T1, HPU for spjeld til utløp, hydraulic schematic for the offer, 09.04.2025 (superseded)
[JB]	Servi 18068 rev. A3, El interface for spjeldventiler (JB1 and operation panel ST-16183), 16.10.2026
[QNA]	QnA Servi hydraulikksystem CAS, meeting notes 18.02.2026
[MAIL]	E-mail thread "S6377070 / 25031 Automatisk stenging", 28.09.–19.10.2026 (Servi, Østerbø, ScaleAQ)
[QG]	HySense QG100 gear flow meter, data sheet and TED
[TDS]	Hoyer HMA3 90L-4 1.5 kW motor data sheet; Marzocchi GHP gear pump data sheet
[HMD]	Hemomatik HMDHI level / temperature switch data sheet
[PID]	CAS P&ID and loop diagrams: CAS-PID-500 (outlet), LD-806 (loop), I/O list, open points
[DB]	CAS design basis (System Modelling) and the outlet simulator CasOutletSim (`design_basis.toml`)
[PLC]	Heimdall PLC export: `FB.Outlet.OutletControl`, DUT `System.Outlet`, `FB.RIO.CommonRIO`, PFC200_Sim `ConfigValues`
[AKS]	Aksjonsliste CAS styresystem (items 23, 41, 42, 81, 82, 91); CAS PLS aksjonsliste (item 6)

1.4 Abbreviations and conventions

HPU	Hydraulic power unit (tank, pump, motor, valves)	PSL	HAWE proportional directional valve, load sensing
JB1	Servi junction box 18068 at the HPU	DOL	Direct-on-line motor starter
CommonRIO	WAGO 750-362 remote I/O node in the CAS cabinet	DO / DI / AI	Digital out / digital in / analog in
Plus stroke	Rod out, oil into the A (piston) side	Minus stroke	Rod in, oil into the B (rod) side
O-35 … S-75	Operating modes (O = operation, S = service; number = residence time in minutes)	FAT	Factory acceptance test

Tags follow the P&ID set [PID] (ISA-5.1), printed as **PT61**. Servi's own designations from [A1] and [JB] are given in brackets, e.g. **PT61** (1PT1). Tags marked † are proposed in this document and are not yet on the P&ID set. PLC names are printed as `Outlet.Cylinder[i].Status.Position`; cylinder index `i` = 0..3 is cylinder 1..4. Open issues are marked **OIn** and listed in Section 6; decisions are marked **DECn**. TBD marks a number that is not known yet.

1.5 Revision

Rev.	Date	Change
A	21.10.2026	First issue. Written from [A1], [JB] rev. A3, [FD] rev. B and the PLC export, after the review of 21.10.2026. Records the power-loss decision DEC1. Supersedes [FD] rev. B §3.2.3 on fail-to-close and XV65.
A1	03.11.2026	CommonRIO link loss also detected by stale data (freshness watchdog) with an automatic reconnect (FR-20).

2 Block diagram description

2.1 Outlet arrangement

The pen water leaves through a bottom centre outlet (Ø 5.5 m [PID]). Four dampers sit in the outlet; together they set the outlet conductance and thus, with the inlet pumps, the water exchange and the over-height of the water in the pen. Each damper is moved by its own hydraulic cylinder through a mechanical link (Østerbø design). In normal operation all four dampers have the same opening, given by the operating mode [FD]. Each damper can also be moved on its own (manual mode), for calibration and maintenance. Below the dampers, branches go to the dead fish and sludge systems, which are not part of this description.

2.2 Equipment

Item	Description	Source
Cylinders 1–4	Double-acting, Ø40/20×300 mm, Servi part 810872, type HRC-N-D-40/20x300-S (Servi A1 item 29). Piston area 12.57 cm^2, annulus 9.42 cm^2. Oil volume per full stroke: A side 0.377 l, B side 0.283 l. Acid-proof rod, double scrapers recommended.	[A1], [QNA]
Tank **TL61**	12 l oil tank (Servi item 1), sight glass, breather filter.	[A1]
Pump **PU61**	Fixed-displacement gear pump Marzocchi GHP1D4, 2.8 cm^3/rev; 3.5 l/min at 150 bar (4.1 l/min theoretical at 1450 rpm).	[A1], [TDS]
Motor **M61** (1M1)	Hoyer HMA3 90L-4, 1.5 kW, 230/400 V 50 Hz, 3.4 A, IE3, with PTC. DOL via KM1, motor protection breaker QM1.	[A1], [TDS], [JB]
Hand pump **HP61**†	Hansa GL25 hand pump with check valve into the pressure line (Servi items 25–26). Manual override when the motor cannot run.	[A1]
Pressure relief	Main relief 150 bar. Max. working pressure 150 bar, design 170 bar, test 190 bar.	[A1]
Valve bank	HAWE PSL size 2, four sections (Servi item 21, 810855), load sensing, one section per cylinder. Two solenoids per section: **XV61**..**XV64** = 1YV1..1YV4, solenoids MB1–MB8 (24 V DC, 680 mA). Per section: port pressure limiters 100 bar, over-centre (load-holding) valves 130 bar on A and B.	[A1], [JB]
Flow meters **FT61**..**FT64** (1FT1..4)	HySense QG100, 0.05–5 l/min, 0.191 cm^3/pulse (≈5200 pulses/l), square-wave output with direction detection (order no. 3146-02-35.0300). On the A line of each cylinder.	[A1], [QG]
Pressure transmitters **PT61**, **PT63**, **PT65**, **PT67** (1PT1..4)	PT5401, 0–250 bar, 4–20 mA, one per cylinder on the A line. The B-line transmitters **PT62**..**PT68** of [FD] are *not* on [A1] OI1.	[A1], [JB]
Tank switch **LSL61**/**TSH61** (1SL1/1ST1)	Hemomatik HMDHI-OT1: low level (switch at 120 mm) and over-temperature 70 °C, NC contacts in series, potential-free.	[A1], [HMD], [JB]
Filter switches **PDS61**†, **PDS62**†	Pressure filter (6 bar bypass) and return filter (1.75 bar bypass) clogging indicators 1PDS1/1PDS2, 24 V, contacts in series (one signal).	[A1], [JB]
Gauge **PI61**	Manometer Ø63, 0–250 bar, on Minimess test point; 9 further Minimess test points.	[A1]
JB1	Servi junction box ST-18068/JB, IP66: terminals for all signals, main and backup 400 V supply with contactors K1 (main) / K2 (backup, interlocked), motor contactor KM1, motor breaker QM1.	[JB]
Operation panel	Servi ST-16183 at the HPU: emergency stop **ES61** (SE1), Stop (S2), Start with green running lamp (SH3).	[JB]
CommonRIO	WAGO 750-362 node in the CAS cabinet: 2×750-638 counters, 750-455 analog inputs, 750-1405 DI, 750-1515 DO with interposing relays XK01–XK08. Modbus TCP to the PLC.	[PID], [PLC]
PLC	WAGO PFC200 CAS PLC, program `FB.Outlet.OutletControl`, called from `PRG.Main`.	[PLC]
HMI	CAS operator HMI (operating mode, manual / jog, calibration, alarms).	–

Figure 1 shows the physical and network structure. Figure 2 shows the hydraulics for one cylinder in simplified form, with the instruments as ISA bubbles; the other three cylinders are identical.

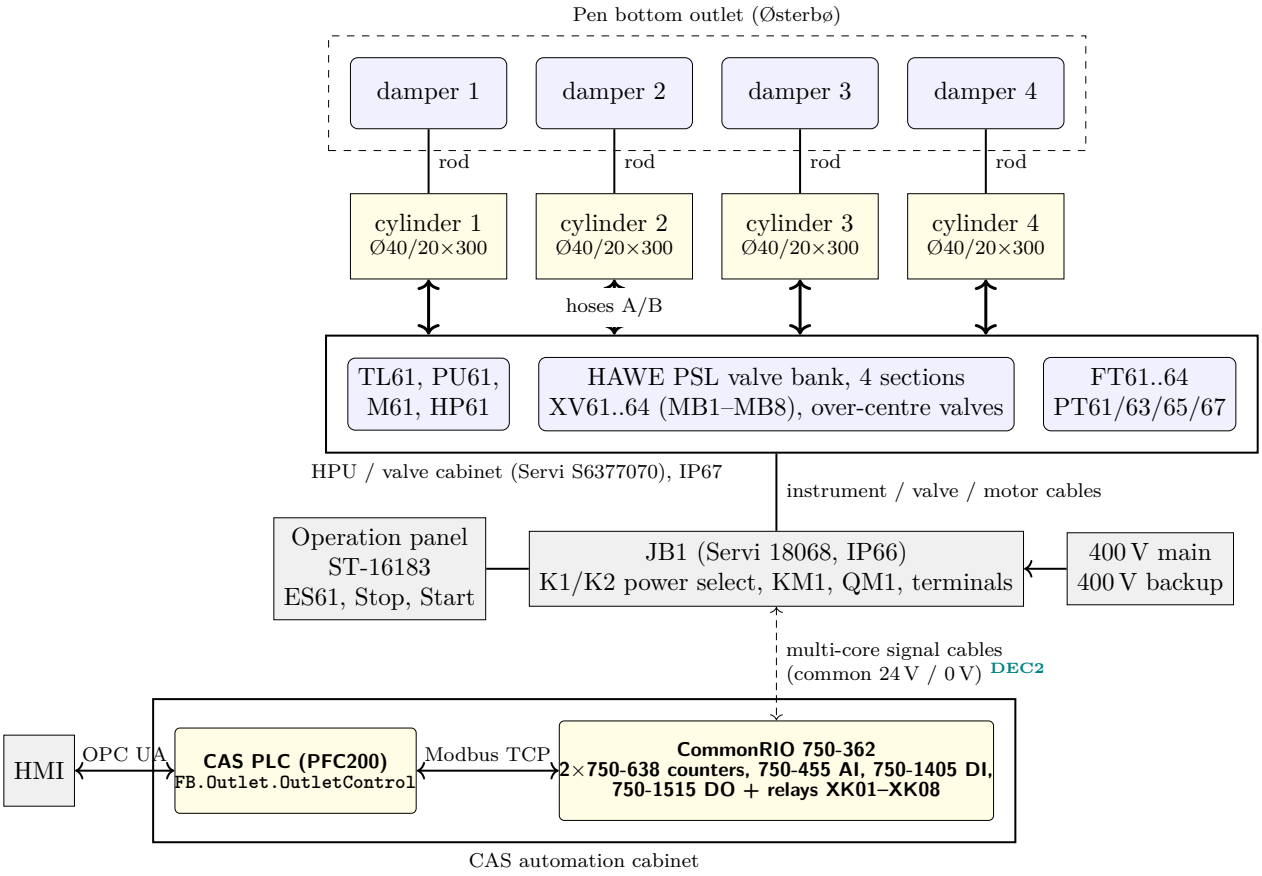


Figure 1: Block diagram of the outlet system: physical arrangement and network. Thick lines are oil hoses and cables; dashed arrows are field signals; solid arrows are network links.

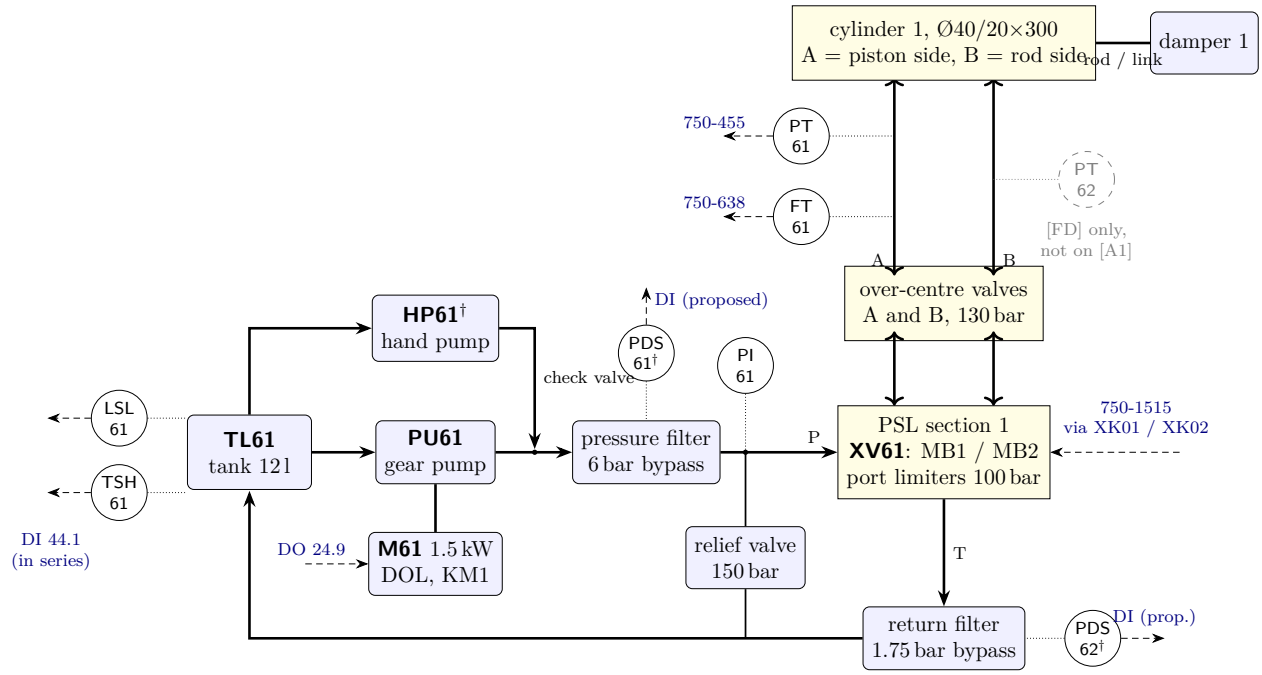


Figure 2: Simplified hydraulic and instrument diagram, cylinder 1 (typical for 1–4). After [A1] and CAS-PID-500. Dotted lines: instrument connection. Dashed arrows: signals to and from CommonRIO, labelled with the module or image address. The valve bank, pump, relief valve, filters and tank are common to all four cylinders.

2.3 How the hydraulics work

The gear pump delivers a fixed flow of about 3.5 l/min into a common pressure line while the motor runs. The PSL valve bank is load sensing: with all sections in neutral, the pump flow returns to the tank at low pressure. When the PLC energises one solenoid of a section, oil flows to the A or B port of that cylinder, and the cylinder moves. The over-centre valves in the A and B lines hold the cylinder where it is when the valve returns to neutral, and keep a pulling load from running the cylinder away. The port pressure limiters (100 bar) cap the pressure on each cylinder port; the main relief valve (150 bar) protects the pump.

All four sections share one pump. When several cylinders move at the same time, the flow divides between them according to their loads: the least-loaded cylinder moves fastest. The total time to move all four through the same stroke is set by the total oil volume divided by the pump flow (Section 5.1).

The oil-flow meter sits on the A line of each cylinder, between the valve bank and the cylinder. It sees the A-side oil volume in both directions: oil in on the plus stroke, oil out on the minus stroke. A full stroke is therefore the same number of pulses both ways (about 1970 pulses, see Table 3). The pressure transmitter of [A1] is on the same A line.

2.4 Signals between the HPU and the PLC

Table 1 lists the signals that JB1 offers [JB] and whether the PLC uses them.

3 Functional diagram description

3.1 Control principle

Figure 3 shows the control chain. It is an on/off position control per cylinder:

1. The **setpoint** is the opening of the operating mode (auto) or a value from the HMI (manual). All four cylinders get the same setpoint in auto; in manual each cylinder can have its own.
2. The **position** of each cylinder is the pulse count from its flow meter, scaled between the calibrated closed and open counts to 0–100 %.
3. The **state machine** (Figure 5) compares position and setpoint, with a hysteresis of 5 %. If the cylinder is outside the band, it asks for oil in the right direction.
4. The **pump** starts when any cylinder asks for oil. After the start-up time (5 s) the solenoid valves are energised. The valve drops when the position reaches the setpoint. The pump stops 2 s after the last request.
5. **End positions** (0 % and 100 %) are not trusted to the pulse count alone. A move to 0 or 100 % continues until the line pressure is above the end-position limit for 4 s, which shows that the piston is against its stop.

3.2 Position measurement

Counting. Each flow meter gives one pulse per 0.191 cm^3 of oil. A WAGO 750-638 counter channel counts the pulses. The counter has no direction input that the PLC uses: the PLC sets the counting direction in the module's control byte from the valve it energises (open valve: count up, close valve: count down). The direction stays as last commanded while the cylinder is idle. The PLC checks that the module reports the commanded direction; a mismatch longer than 2 s is a fault (`DirectionOK` = FALSE). The QG100 ordered has its own direction output, which is not used [OI10].

Table 1: Signals at JB1 (Servi 18068 rev. A3) and their use in the PLC.

Signal [JB]	Terminal	Type	PLC (current export)
Level / temperature switch tank (LSL61/TSH61)	X1:1–2	DI, NC	`Outlet.Input.HydraulicOilTankFault`, DI 44.1. Alarm only OI9.
Pressure transmitter cyl. 1–4	X1:3–12	AI 4–20 mA	`PressureOpen`: AI words 28/30/32/34. `PressureClose` words 29/31/33/35 have no transmitter OI1.
Filter clogged (PDS61 + PDS62 in series)	X1:13–20	DI	Not mapped OI7.
Flow meter cyl. 1–4	X2:1–12	pulse (A2: corrected from 4–20 mA)	750-638 counters, words 12–19 OI10.
Valve open / close cyl. 1–4 (MB1–MB8)	X2:13–24	DO 24 V via relays	DO 24.0–24.7 (`OpenValve`/`CloseValve`) OI12.
Activate main power (K1) / activate backup power (K2)	X1:21–22	DO (from PLC)	Not mapped OI6.
Activate motor (KM1)	X1:23	DO, potential-free	`HydraulicPump.Output.Start`, DO 24.9.
Motor running (KM1 aux)	X1:24	DI	Not mapped; PLC shows commanded state OI7.
Motor overload (QM1 97/98)	X1:25	DI	`HydraulicPump.Input.StarterFault`, DI 44.0 (assumed to be this contact) OI7.
Emergency stop pressed / E-stop loop	X1:26–35	DI	`GVLMain.EmergencyStop` = NOT DI 44.2. No alarm OI8.
Local Start / Stop (ST-16183)	X1:26–35	DI	Not mapped OI16.
Power lost / pump on backup supply	–	DI	Not on [JB]; requested in [AKS] item 82 OI6.
–	–	DO	PLC also has `HydraulicPump.Output.Reset` (DO 24.10). QM1 is reset by hand; no remote reset on [JB] OI7.

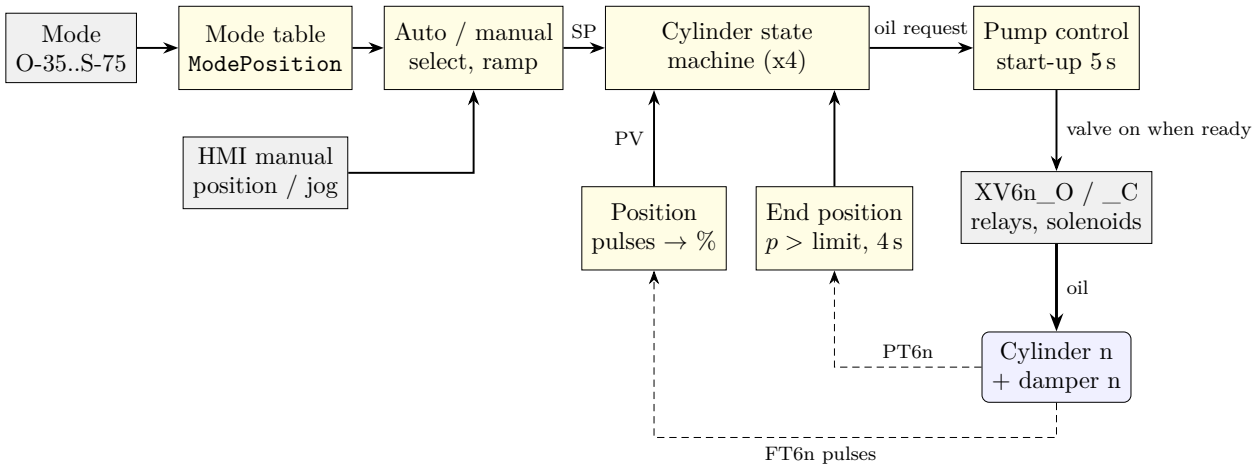


Figure 3: Control principle of one damper (functional diagram after [COO, §14.3.3]). Yellow: PLC functions; grey: operator and actuators; blue: process.

Scaling.

$$\text{Position}_n\,[\%] = \frac{\text{Counter}_n - \text{ClosedPosition}_n}{\text{OpenPosition}_n - \text{ClosedPosition}_n} \cdot 100$$

`ClosedPosition` and `OpenPosition` are retained per cylinder (`CylinderConfigDUT`; defaults 0 and 2000 pulses). The position is only valid (`Calibrated` = TRUE) when the span is positive, the remote I/O link has been up for 1 s, no counter set is in progress, and the direction is confirmed. The outlet position shown on the HMI is the mean of the four cylinders.

Calibration (homing). [FD] and the PLC use the end stops as reference:

- **Define closed position** (`Command.CalibrateClosed`): the PLC writes 0 into the counter with the 750-638 set-counter handshake and sets `ClosedPosition` = 0. The command is accepted in any state; the operator must first drive the damper fully closed.
- **Define open position** (`Command.CalibrateOpen`): accepted only in state `Opened`, i.e. after the open end position is confirmed by pressure. The current count is stored as `OpenPosition`.
- **Check at every end position**: when an end position is confirmed by pressure, the PLC compares the counted position with 0 or 100 %. A deviation larger than the calibration tolerance (5 %) raises *calibration error*. So does a position outside −5 . . . 105 %.

Servi recommends running the cylinders to their end positions once or twice a week, to keep the rods free of fouling and to re-check the calibration [QNA]. The PLC has no automatic exercise function; it is an operator routine **OI17**.

Counter restore. The 750-638 restarts at 0 after it loses power. The PLC keeps the last good count in retain memory. If the module suddenly shows about 0 (within 5 pulses) while the retained count is more than 50 pulses away, the PLC writes the retained count back into the module. The position is held invalid until the module confirms; no confirmation within 3 s raises a flow sensor fault.

Drift. If the count changes by more than 2 pulses while no valve has been energised for 0.5 s, the PLC sets `PositionDrift` (flow sensor fault). The limit 2 is from [FD], marked HOLD there **OI18**.

3.3 End-position detection

When the piston reaches its stop, the oil flow stops and the line pressure rises to the port limiter setting. The PLC sees an end position when the filtered pressure on the driven side is above `MaxPressureOpen` (or `MaxPressureClose`) for `MaxPressureOpenDelay` (4 s) while the valve for that direction is on. The delay filters short pressure peaks at valve switching [QNA].

Two hardware facts limit this today:

- **Only one transmitter per cylinder.** [A1] has one PT per cylinder, on the A line. The PLC and [FD] expect one on each side. With the A side only, the plus-stroke end can be detected but the minus-stroke end cannot. With the PLC defaults, a command to 0 % then runs until the travel timeout (30 s) and ends in a fault, because the state machine waits for close-side pressure that never comes **OI1**. Which stroke is “open” must be confirmed **OI2**.
- **Limit equal to the port limiter.** The default limit is 100 bar and the test is “greater than”. The port pressure limiters of [A1] are also set to 100 bar, so the pressure may never exceed the limit **OI3**.

The alternative to a second transmitter is to detect the end of stroke by the pulses stopping while the valve is on (Servi’s suggestion in [QNA]). The PLC does not implement this.

3.4 Operating modes

The operating mode sets the water flow through the pen. The dampers get a fixed opening per mode, and the level controller LIC01 gets the level setpoint for the mode [FD Tabell 10]. The mode is chosen on the HMI (`Outlet.Command.Mode`) and kept over a restart (`GVLRetain.OutletMode`).

Table 2: Operating modes and damper openings.

Mode	Residence [min]	Water flow [m^3/h]	Level SP [cm]	Opening, PLC default [%]	Comment
O-35	35	34 000	6	20	`OutletValveMode.O35` = 0
O-45	45	27 000	2.5	30	`O45` = 1
O-60	60	20 000	0.5	40	`O60` = 2
S-75	75	16 000	0.5	50	`S75` = 3, service

The opening column of [FD] is empty. The PLC defaults (`ModePosition` = [20, 30, 40, 50] %) *increase* while the flow *decreases*. If 100 % means open, this looks reversed. The values, and the relation between cylinder stroke and damper opening, must come from Østerbø's damper design or from tests [OI5]. The openings can be changed on the HMI and are retained.

3.5 Auto, manual and jog

Outlet auto
: All four cylinders follow the mode opening. Default after first start.

Outlet manual
: All four cylinders follow a common manual position from the HMI. The switch is bumpless: the manual position starts at the setpoint that was active in auto.

Cylinder manual
: One cylinder follows its own manual setpoint (e.g. for calibration). The others stay in auto. Bumpless as above.

Jog
: In cylinder manual, the operator holds *open* or *close*. The valve stays on while the command is held. Jog does not need a valid position (so it can be used before calibration). Jog stops at a pressure-confirmed end position, and after `MaxTravelTimeOpen/Close` (30 s) of valve-on time; the limit holds until the command is released. On release the manual setpoint is set to the actual position, so the cylinder stays where it is.

`RampRateOpen` / `RampRateClose` can rate-limit the setpoint (%/s); the default 0 means no ramp.

3.6 Sequences

Move (open or close). Figure 4 shows a move of one or more cylinders.

1. A cylinder's position leaves the band setpoint ±5 %, or the setpoint is 0 or 100 % and the end position is not confirmed. The state machine goes to `Opening` or `Closing` and requests oil.
2. The PLC starts the pump (DO 24.9). Valves stay off for `PumpStartUpTime` (5 s).
3. The valve of the requested direction is energised (**XV6n**_O or _C). The counter direction follows in the same Modbus telegram.
4. The valve drops when the position reaches the setpoint, or, for 0 / 100 %, when the end position is confirmed by pressure (4 s).
5. The pump stops `PumpStopDelay` (2 s) after the last request, to avoid a restart between moves.

Stop. A move stops when the target is reached, on emergency stop, on a cylinder fault, or when the position becomes invalid. A stopped cylinder holds its position mechanically on the over-centre valves. There is no separate operator "stop" command for an auto move; setting the outlet to manual at the present position has the same effect.

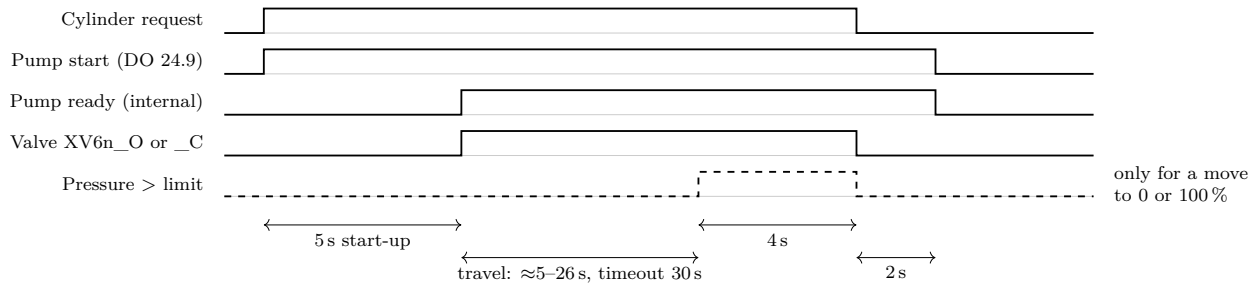


Figure 4: Timing of a move. For a move to an intermediate position the valve drops when the position reaches the setpoint, without the pressure step.

3.7 Pump control

- **Auto** (default): the pump runs while any cylinder requests oil, plus 2 s.
- **Manual**: the operator starts and stops the pump from the HMI (`HydraulicPump.Command.Start/Stop`).
- The start output is blocked by emergency stop and by starter fault (`Status.Available` = NOT `StarterFault`).
- `Status.Running` is the commanded state; there is no running feedback in the PLC **OI7**.
- The local panel at the HPU can also start and stop the motor [JB]. How local and remote start are combined is not defined **OI16**.

3.8 Cylinder state machine

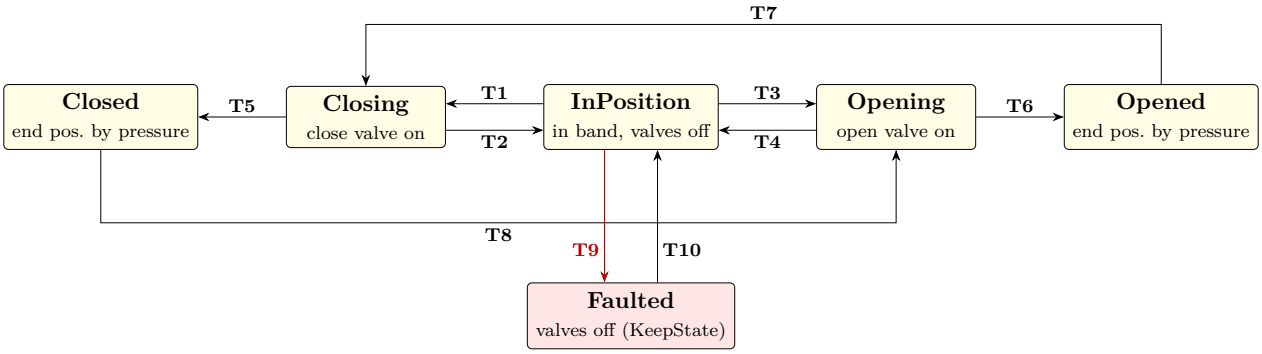
Figure 5 shows the states of one cylinder as implemented in the method `CylinderStateMachine` of `OutletControl` (enum `ValveState`). The direction outputs follow the state: `Opening` requests the open valve, `Closing` the close valve; all other states request nothing, except jog in manual. "Blocked" means the cylinder has a timeout fault or no valid position.

Fault action. The PLC calls the state machine with `FaultActionMode.KeepState`: in `Faulted` both valves are off and the cylinder stays where it is. The modes `Close` and `Open` exist in the code but are not used. This matches the power-loss decision **DEC1**: the outlet is never moved automatically to a safe end position.

Fault recovery. The timeout fault is the alarm state of the travel-time alarm. When the valve drops, the timer resets; when the operator acknowledges, the alarm returns to normal and the cylinder leaves `Faulted`. From `InPosition` it will try again if the position is still outside the band. `Command.FaultReset` exists in the DUT but is not used.

4 Functional requirements

Requirements are numbered FR-n. "Impl." marks requirements the current PLC export meets.



T	Condition (hyst. = `PositionCounterHysteresis`, SP = ramped setpoint)
T1	pos > SP + hyst., or SP = 0 and pos > 0 (a move to 0 % always ends on pressure)
T2	pos ≤ SP and SP > 0
T3	pos < SP − hyst., or SP = 100 and pos < 100
T4	pos ≥ SP and SP < 100
T5 / T6	End position confirmed: driven-side pressure > limit for 4 s
T7	pos > SP + hyst. and SP < 100
T8	pos < SP − hyst. and SP > 0
T9	Blocked: timeout fault, or no valid position. Applies to every state, drawn for InPosition only
T10	Not blocked any more; InPosition then picks the direction

Figure 5: State diagram of one cylinder (`ValveState`; the transient `MovingToPosition` goes straight to `InPosition`). Emergency stop does not change the state; it switches both valves off and freezes the ramped setpoint at the actual position.

4.1 Processes

FR-1 The system shall hold all four dampers at the opening of the selected operating mode (auto). Impl.

FR-2 The operator shall be able to set a common manual opening, and to put each cylinder in manual with its own setpoint and jog. Impl.

FR-3 The position of each cylinder shall be computed from flow meter pulses, 0–100 %. Impl.

FR-4 Moves to 0 and 100 % shall end on a pressure-confirmed end position. Impl. for the side with a transmitter OI1

FR-5 The pump shall run only while a cylinder needs oil (auto), or on operator command (manual). Impl.

FR-6 The operator shall be able to calibrate closed and open positions per cylinder, and the PLC shall check the calibration at every end position. Impl.

4.2 Inputs and outputs

The inputs are the four pulse counters, the pressure transmitters, the tank switch, the starter fault and the emergency stop; the outputs are eight solenoid valves and the pump start. The full list with addresses is Table 7; ranges and resolution are in Table 6.

FR-7 All JB1 signals that the operator needs to act on shall be wired to the PLC: filter clogged, motor running, motor overload, E-stop pressed, power supply state OI6 OI7.

FR-8 The pressure transmitters shall be scaled 0–250 bar in the retained configuration. The DUT default is 0–10; the simulator `ConfigValues` set 0–250 OI4.

4.3 Start-up and shut-down

FR-9 After a PLC restart the mode, the mode table, the calibration (closed/open counts), the last counter value and the transmitter configuration shall be restored from retain memory. Impl.

FR-10 After a restart or a loss of the remote I/O link, no cylinder shall move until the link has been up for 1 s and the counter values are valid. If a counter module has restarted at 0, the retained count shall be written back first. Impl.

FR-11 A restart shall not move the dampers by itself, except to bring a cylinder that is outside the hysteresis band back to its setpoint. Impl.

FR-12 First commissioning: each cylinder shall be calibrated (close fully, define closed; open fully, define open) before auto is used. The default span of 2000 pulses is treated as calibrated by the PLC, so this relies on the commissioning procedure OI15.

FR-13 Shut-down: on emergency stop the pump stops and all valves drop. The dampers stay where they are. Impl.

4.4 Power loss

Decision DEC1 (21.10.2026): the outlet does *not* close automatically on power loss.

- The dampers hold their position: the over-centre valves lock the cylinders, and the PLC fault action is `KeepState`.
- If the outlet must be moved without power, the operator does it by hand at the HPU: the hand pump (**HP61**†) gives pressure, and the valve section is shifted by its manual actuator OI17.
- An accumulator for automatic closing is deferred. The HPU cabinet has room to retrofit one later if needed.
- JB1 has a backup 400 V supply (K2) for the pump; how it is selected is open OI6.

This supersedes [FD] rev. B §3.2.3 “Utløpet skal stenge ved strømbortfall” (fail to close) and the shut-off valve **XV65**. XV65 is not on [A1]; its DO (24.8) and cable are spare.

FR-14 A “power lost / pump on backup supply” signal from the HPU (relay contact, [AKS] item 82) shall reach the PLC, and the operator shall get an alarm OI6.

4.5 Fault identification and recovery

Table 4 lists the alarms. Common rules:

FR-15 A cylinder whose valve is on for longer than the travel time shall stop and give an alarm (*long opening / closing time*). It stays stopped until the alarm is acknowledged. Impl.

FR-16 Count changes without an active valve, a wrong count direction, or a counter set that is not confirmed shall give a *position measurement fault*. Impl.

FR-17 A deviation between counted and pressure-confirmed end position larger than the tolerance shall give a *calibration* warning. Impl.

FR-18 Low oil level or high oil temperature shall give an alarm. Whether it shall also stop the pump is open OI9.

FR-19 Emergency stop, clogged filter and motor overload shall each give an alarm OI7OI8.

4.6 Communication

FR-20 The PLC reads and writes the CommonRIO image over Modbus TCP every cycle. A lost link gives the alarm *CommonRIO communication error*; the cylinder positions become invalid and the cylinders go to `Faulted` (valves off). The link also counts as lost when the

connection is open without error but no new response has come for 2 s (freshness watchdog: a marker in the spare input word 27 that every response overwrites; `Status.DataStale`); the connection is then closed for 1 s and opened again, at once and every 10 s while the data stays stale (`Status.Restarts`). Impl.

FR-21 On a lost link the remote I/O shall switch its outputs (valves, pump start) off, so the dampers hold. The behaviour of the 750-362 fieldbus watchdog is not set, and [FD] Tabell 4 is empty OI13.

FR-22 The HMI shall show per cylinder: position, setpoint, state, valve outputs, line pressure, calibration status and alarms; and for the HPU: pump state, tank and filter status, power supply state.

5 Non-functional requirements

5.1 Timing and capacity

Table 3 gives estimates from the cylinder data and the pump flow. [FD] leaves the stroke time open ("ca nn sekunder"); these numbers should replace it after the FAT.

Table 3: Estimated stroke times, pulse counts and pulse rates.

Quantity	Basis	Value
Oil volume, plus stroke (A side)	$\pi \cdot 2.0^2\,\text{cm}^2 \cdot 30\,\text{cm}$	0.377 l
Oil volume, minus stroke (B side)	$\pi(2.0^2 - 1.0^2)\,\text{cm}^2 \cdot 30\,\text{cm}$	0.283 l
Pulses per full stroke (A line, both directions)	$377\,\text{cm}^3/0.191\,\text{cm}^3$	≈1970
[FD] / PLC default span	0.4 l × 5200 / default `OpenPosition`	2080 / 2000
Position resolution	1 pulse	0.05 % ≈ 0.15 mm
Stroke time, one cylinder alone, plus / minus	0.377 l and 0.283 l at 3.5 l/min	6.5 s / 4.9 s
Stroke time, four cylinders together, plus / minus	1.51 l and 1.13 l at 3.5 l/min	26 s / 19 s
add pump start-up	`PumpStartUpTime`	5 s
Travel timeout	`MaxTravelTimeOpen/Close`	30 s
Pulse rate, all four moving	≈0.9 l/min per cylinder	≈75 Hz
Pulse rate, one cylinder, plus stroke	3.5 l/min (4.1 l/min unloaded)	300 Hz (355 Hz)
Pulse rate, one cylinder, minus stroke	A-side outflow = pump flow × 12.57/9.42 = 4.7 (5.4) l/min	410 Hz (470 Hz)
750-638 max. input frequency	data sheet	500 Hz
QG100 measuring range	data sheet	0.05–5 l/min

Findings:

- A full plus stroke of all four cylinders takes about 26 s of valve-on time. The travel timeout of 30 s leaves about 4 s (15 %). Cold oil, wear and the over-centre back pressure can use that up, and the move then ends in a fault. The timeout should be set from FAT measurements, e.g. 1.5× the measured time OI4.
- The design basis and simulator assume 30 s per stroke and a 160 bar relief [DB]; they should be updated to the numbers above and to [A1] OI3.
- A single cylinder on the minus stroke pushes about 4.7–5.4 l/min out through its flow meter: at or above the top of the QG100 range and close to the 500 Hz of the 750-638. Jog of one cylinder on the minus stroke is the worst case. Check the count at the FAT OI11.
- The QG100 is calibrated in the arrow direction; accuracy in reverse flow (minus stroke) is not

stated OI11. The end-position check catches a drift, but only at end positions.

Other timing: PLC cycle and Modbus TCP update are fast compared with the pulse period at the setpoint (300 Hz → 3 ms per pulse; 5 % hysteresis = ≈100 pulses), so the overshoot after the valve drops is small compared with the hysteresis. Pressure end-position delay 4 s; direction check delay 2 s; counter-set timeout 3 s; comm valid delay 1 s.

5.2 Pressure thresholds

Setting	Value	Source / comment
Main relief valve	150 bar	[A1]; design 170, test 190 bar
Over-centre valves, A and B	130 bar	[A1] ([T1]: 110 bar)
Port pressure limiters	100 bar	[A1]
PLC end-position limit `MaxPressureOpen/Close`	100 bar	`CylinderConfigDUT` default; must be below the port limiter, e.g. 70–80 bar, and above the running pressure (TBD, measure at FAT) OI3
End-position delay	4 s	`MaxPressureOpen/CloseDelay`
Transmitter range	0–250 bar	PT5401; PLC scaling must match OI4

5.3 Alarms

The flow deviation alarms of [FD] Tabell 11 (estimated inflow too low / too high) belong to the inlet and level function and are not repeated here.

5.4 Parameters

All parameters below are retained in `GVLRetain` (outlet configuration, cylinder configuration, transmitter configuration and last counter value per cylinder) and can be changed on the HMI. The values are the DUT defaults; production values are TBD until the FAT.

Table 5: Outlet parameters (`OutletConfigDUT`, `CylinderConfigDUT`) and fixed constants.

Parameter	Default	Meaning
`ModePosition[O35..S75]`	20, 30, 40, 50 %	Opening per mode OI5
`Mode`	O35	Selected mode (retained)
`MaxPressureOpenDelay` / `CloseDelay`	4 s	Pressure above limit this long = end position
`MaxTravelTimeOpen` / `Close`	30 s	Travel timeout; also jog limit (0 = no jog limit) OI4
`RampRateOpen` / `Close`	0 %/s	Setpoint ramp, 0 = off
`PositionCounterHysteresis`	5 %	Dead band around the setpoint
`CalibrationTolerance`	5 %	Calibration check at end positions (min. 0.5 %)
`PumpStartUpTime`	5 s	Pump runs this long before valves open ([FD]: 5 s)
`PumpStopDelay`	2 s	Pump keeps running after the last request
`EnableTimeoutAlarms`	TRUE	FALSE suppresses the travel timeout alarms *and* the fault
`ClosedPosition` (per cyl.)	0	Counter value at closed

Parameter	Default	Meaning
`OpenPosition` (per cyl.)	2000	Counter value at open (calibrated ≈1970)
`MaxPressureOpen` / `Close` (per cyl.)	100 bar	End-position pressure limit OI3
`PressureOpen/Close.Config` (per cyl.)	0–10 (DUT); 0–250 (sim)	Transmitter scaling OI4
`DIRECTION_DELAY`	2 s	Constant: direction mismatch tolerated
`DRIFT_LIMIT`	2 pulses	Constant: drift without valve ([FD], HOLD) OI18
`ZERO_BAND` / `JUMP_LIMIT`	5 / 50 pulses	Constant: counter restart detection
`COUNTER_SET_TIMEOUT`	3 s	Constant: set-counter handshake
Fault action	KeepState	Constant in the call; matches DEC1

5.5 Instruments

Table 6: Outlet instruments and actuators (typical cylinder n = 1..4; tags per [PID]).

Tag	Function	Location	Signal / interface	Range	Resolution	Use
FT61.. **FT64**	Oil flow to/from A side, position by pulse count	HPU cabinet, A line	Pulse 24 V, 3-wire, to 750-638	0.05–5 l/min	0.191 cm^3 (0.05 %)	Control
PT61, **PT63**, **PT65**, **PT67**	Pressure A side (plus stroke), end position	HPU cabinet, A line	4–20 mA to 750-455	0–250 bar	≈0.12 bar	Control, alarm
PT62, **PT64**, **PT66**, **PT68**	Pressure B side (minus stroke)	–	4–20 mA, AI reserved	0–250 bar	–	Not installed OI1
XV61.. **XV64**	Directional valve, _O / _C solenoids (MB1–MB8)	HPU valve bank	DO 750-1515 via relays XK, 24 V 680 mA	on / off	–	Control
PU61/ **M61**	Hydraulic pump and motor	HPU	DO start (KM1), DI overload	3.5 l/min, 150 bar	–	Control
LSL61/ **TSH61**	Tank low level / oil temperature high	HPU tank	DI, NC, series	120 mm / 70 °C	–	Alarm
PDS61†/ **PDS62**†	Pressure / return filter clogged	HPU	DI, series	5 / 1.5 bar Δp	–	Alarm (prop.)
ES61	Emergency stop	Panel ST-16183	DI, NC loop	–	–	Interlock
PI61	Pressure gauge	HPU	local	0–250 bar	–	Local
ZI61.. **ZI64**	Damper position (computed)	PLC	–	0–100 %	0.05 %	HMI

5.6 I/O list

Table 4: Outlet alarms. Status: Impl. = in the PLC export; FD = in [FD]/[AL] but not in the PLC; Prop. = proposed here.

Alarm	Condition	Delay / limit	Action	Status
Long opening time, cyl. n	Open valve on longer than limit	30 s (travel time open)	Valve off, cylinder faulted until ack.	Impl.
Long closing time, cyl. n	Close valve on longer than limit	30 s (travel time close)	As above	Impl.
Position measurement fault, cyl. n (`FlowSensorFault`)	Drift > 2 pulses without valve; or counting direction wrong; or counter set not confirmed	0.5 s / 2 s / 3 s	Alarm; position invalid while direction wrong	Impl.
Calibration error, cyl. n	$\lvert\text{counted} - \text{end position}\rvert >$ tolerance at a confirmed end, or position outside $-5 \ldots 105\,\%$	5 % (calibration tolerance)	Warning; recalibrate	Impl.
Pressure transmitter fault, cyl. n	Wire break / out of range (AnalogSensor)	module status	Alarm	Impl.
Low level / high temperature, hydraulic tank	LSL61 or TSH61 (series)	none in PLC; [FD]: configurable	Alarm only **OI9**	Impl.
Hydraulic pump starter fault	DI 44.0 (motor overload QM1)	–	Pump not available	Impl.
CommonRIO communication error	Modbus TCP link down	–	Positions invalid, valves off	Impl.
Emergency stop activated	ES61	–	Pump and valves off (done); alarm missing	FD **OI8**
Filter clogged	PDS61/PDS62	TBD	Alarm, change filter	Prop. **OI7**
Power lost / pump on backup	relay from HPU	TBD	Alarm	Prop. **OI6**
Motor not running	start on, no running feedback	TBD	Alarm	Prop. **OI7**

Table 7: Outlet I/O on CommonRIO (Modbus TCP image, [PLC] `FB.RIO.CommonRIO`; modules per [PID]).

Module / ch.	Image	JB1 terminal	PLC variable
Inputs			
-R1-004 750-638 ch1/2	IN 12/13, 14/15	X2:1–6	`Outlet.Cylinder[0..1].Input.OilFlowStatusWord` / `OilFlowCounter`
-R1-005 750-638 ch1/2	IN 16/17, 18/19	X2:7–12	`Outlet.Cylinder[2..3].Input...`
-R1-009 750-455 ch1–4	IN 28–31	X1:3–7	`Cylinder[0].PressureOpen` (28), `Cylinder[0].PressureClose` (29, no PT), `Cylinder[1]` (30, 31)
-R1-010 750-455 ch1–4	IN 32–35	X1:8–12	`Cylinder[2]` (32, 33), `Cylinder[3]` (34, 35); odd words have no PT OI1
-R1-017 750-1405 ch1	IN 44.0	X1:25	`HydraulicPump.Input.StarterFault`
-R1-017 750-1405 ch2	IN 44.1	X1:1–2	`Outlet.Input.HydraulicOilTankFault`
-R1-017 750-1405 ch3	IN 44.2	X1:26–35	`GVLMain.EmergencyStop` := NOT 44.2 (NC loop)
-R1-017 750-1405 ch4..	IN 44.3..	X1:13–24, 26–35	Spare: filter, motor running, local start/stop, power state OI6 OI7
Outputs			
-R1-004/005 750-638	OUT 12–19	–	`OilFlowControlWord` / `OilFlowCounterSet` per cylinder
-R1-019 750-1515 ch1–8	OUT 24.0–24.7	X2:13–24	`Cylinder[0..3].Output.OpenValve` / `CloseValve`, via relays XK01–XK08
-R1-020 750-1515 ch1	OUT 24.8	–	Spare (was XV65, superseded DEC1)
-R1-020 750-1515 ch2	OUT 24.9	X1:23	`HydraulicPump.Output.Start`
-R1-020 750-1515 ch3	OUT 24.10	–	`HydraulicPump.Output.Reset`: no counterpart OI7
-R1-020 750-1515 ch4..	OUT 24.11..	X1:21–22	Spare: activate main / backup supply OI6

Cabinet drawing 792.001 rev. 1 still shows DI module -R1-017 and DO -R1-020 ch2/3 as spare; they must be wired for the PLC mapping above [PID]. The header comment in `CommonRIO.OutputMapping` names the DO modules -R1-018/019 and "750-455"; the P&ID set has -R1-019/020 750-1515.

5.7 HMI operator functions

The HMI screens for the outlet are not built yet OI19. The PLC offers these commands:

Function	PLC interface
Select operating mode	`Outlet.Command.Mode` (O35..S75; invalid values ignored)
Edit mode openings	`Outlet.Config.ModePosition[]`
Outlet auto / manual, manual position	`Outlet.Command.Auto/Manual`, `ManualPosition`
Cylinder auto / manual, setpoint	`Outlet.Cylinder[i].Command.Auto/Manual`, `ManualSetpoint`
Jog open / close (hold to run)	`Outlet.Cylinder[i].Command.Open/Close`
Define closed / open position	`Outlet.Cylinder[i].Command.CalibrateClosed/CalibrateOpen`
Pump auto / manual, start / stop	`Outlet.HydraulicPump.Command.Auto/Manual/Start/Stop`
Acknowledge alarms	`Outlet.Cylinder[i].AlarmAck.*`, `Outlet.AlarmAck.*`, `HydraulicPump.AlarmAck.*`
Show	position, setpoint, target, state, valve outputs, pressures, calibration deviation, pump state

Some outlet commands exist in the DUT but are not used by the code: `Open`, `Close`, `FaultReset`, `ManualPumpStart` and `ManualPumpStop`.

5.8 Verification

The outlet simulator CasOutletSim (heimdall-cas-simulation) models piston, pulses and line pressures, and injects jammed cylinder, counter drift, transmitter faults and a pump that runs without pressure [DB]. The integrated FAT with the Servi cabinet at Østerbø is planned for the end of October 2026. It should at least measure: stroke times (single and four together, both directions), running and end-stop pressures, pulses per full stroke both ways, drift with the valves in neutral, the minus-stroke pulse rate, the valve response to on/off relay drive, and the emergency-stop and power-loss behaviour **OI20**.

6 Open issues and decisions

6.1 Decisions

No.	Date	Decision
DEC1	21.10.2026	The outlet does not close automatically on power loss. Dampers hold position (over-centre valves, PLC `FaultActionMode.KeepState`); manual override with the hand pump at the HPU. Accumulator deferred; there is space in the cabinet to retrofit it. Supersedes [FD] rev. B fail-to-close and XV65.
DEC2	16.10.2026	One multi-core signal cable with common 24 V / 0 V, bridged in JB1; separate conductors for the valve coils [MAIL].

6.2 Open issues

No.	Issue	Owner / next step
OI1	**One pressure transmitter per cylinder.** [A1] has 1PT1–1PT4 on the A line only; [FD] (PT61–68), the P&ID and the PLC (8 AIs, `PressureOpen` + `PressureClose`) expect two. The minus-stroke end cannot be confirmed; with `MaxPressureClose` > 0 a move to 0 % ends in a travel timeout. Options: buy a second transmitter on a Minimess point (Servi offered, 19.10, extra cost), or detect end of stroke by pulses stopping (PLC change). Bjørn asked 19.10 who removed 4 of 8.	JP / Bjørn / Servi
OI2	**Plus stroke = open?** PLC and P&ID assume A side / plus stroke opens the damper. Confirm with Østerbø's damper linkage; it decides which end the single PT confirms and whether `CalibrateOpen` is possible.	Østerbø
OI3	**End-position limit vs port limiter.** PLC limit 100 bar (strict $>$) equals the 100 bar port limiters of [A1]; may never trip. Set below (e.g. 70–80 bar) after measuring running pressure. Also align [DB]/simulator (160 bar relief, 40 bar load assumed).	Bjørn / FAT
OI4	**Travel timeout and scaling.** Four cylinders together need ≈26 s; `MaxTravelTime` 30 s has little margin. [FD] "ca nn s" to be filled in. PT scaling DUT default 0–10; production retain must be 0–250 bar.	Bjørn / FAT
OI5	**Mode openings.** [FD] column empty; PLC default [20, 30, 40, 50] % rises while flow falls (looks reversed). Need openings per mode and the stroke → opening relation.	Østerbø / process
OI6	**Power state to the PLC.** "Power lost / pump on backup" relay ([AKS] 82) not on [JB]; K1/K2 "activate main / backup" inputs on [JB] are not mapped. Who selects the backup supply (PLC, PMS, local)? Alarm wanted.	JP / Servi / Bjørn
OI7	**JB1 signals not mapped.** Filter clogged, motor running, motor overload (assumed = `StarterFault`), E-stop pressed / loop. PLC `Reset` output (24.10) has no counterpart (QM1 manual). Wire DI -R1-017 and DO -R1-020.	Bjørn / Knut
OI8	**Missing alarms.** No emergency-stop alarm (ES61 in [AL]); no filter, overload, motor-not-running or power alarms.	Bjørn
OI9	**Oil tank fault only alarms.** Low level / 70 °C does not stop the pump; no delay as [FD] asks. Decide whether to stop the pump (pump protection).	Bjørn / Servi
OI10	**Flow meter signal.** [JB] overview sheet says 4–20 mA, rev. A2 says pulse (PLC assumes pulse); Servi to correct the overview. The direction output of the QG100 (type 46) is unused.	Servi
OI11	**Pulse rate and reverse flow.** One cylinder on the minus stroke: 4.7–5.4 l/min, 410–470 Hz, at the top of the QG100 range and near the 500 Hz 750-638 limit. Reverse-flow accuracy of the QG100 not stated. Test at FAT.	FAT
OI12	**PSL valves on/off.** The PSL sections are proportional; [JB] sheet 8 shows an amplifier option (±10 V, 0.63 A, 4–20 mA setpoint). The PLC drives the 24 V coils (680 mA) on/off through relays XK. Confirm with Servi that full on/off drive is right (min. ≈250 mA [QNA], optional speed card) and the relay contact rating.	Servi / Knut
OI13	**Comm-loss output behaviour.** [FD] Tabell 4 empty; 750-362 watchdog not configured. Proposed: valves and pump start off (hold).	Bjørn

No.	Issue	Owner / next step
OI14	**Data disagree.** Motor 1.5 kW 400 V ([A1], [TDS], [JB]) vs "under 1 kW" ([QNA]); [JB] cable -W1M1 labelled "24Vdc"; pressures 150/130 bar ([A1]) vs 160/110 ([DB], [T1]); tank 12 l ([A1]) vs 10 l in earlier notes; cylinder type also written "HCR-C-D-40/20x300" vs "HRC-N-D-40/20x300" on the [A1] part list. Motor starter location [AKS] 91.	JP / Servi
OI15	**Calibration state.** The default span (0–2000) counts as calibrated; no "never calibrated" state. Calibration must be part of commissioning. `Command.FaultReset` unused.	Bjørn
OI16	**Local / remote.** Local Start / Stop on ST-16183 is not in the PLC; priority between local panel and PLC start, and whether E-stop also removes coil power, not defined.	Servi / Bjørn
OI17	**Manual override and exercise.** Write the hand-pump procedure (which valve to operate, manual levers on the PSL sections to be confirmed) and the weekly end-position exercise into the O&M manual.	Østerbø / Servi
OI18	**Drift limit.** 2 pulses (HOLD in [FD]); check against measured leakage and thermal drift with the over-centre valves.	FAT
OI19	**HMI.** Outlet screens not built; content per this document.	HMI
OI20	**Integrated FAT** at Østerbø (end of October 2026): test plan per the verification section.	Fredrik / JP